UNIT:- 4 DIFFERENTIATIONS

Q① find the second order derivative of $e^{2x} \cdot \cos 3x + x^4$

Q② Obtain the local maxima or local minima of $f(x) = x^3 - 6x^2 + 9x + 15$. Also find the local maximum or local minimum values $f(x)$

Q③ find the differential co-efficient of $\sin^{-1}x$ from the first principle

Q④ If $x^m y^n = (x+y)^{m+n}$ then evaluate $\frac{dy}{dx}$,

Q⑤ prove that $\operatorname{sech}^2 x + \tanh^2 x = 1$

Q⑥ find $\frac{dy}{dx}$ when $y = x^{\sin x} + (\sin x)^x$

Q⑦ If $y = \sqrt{\cos x + \sqrt{\cos x + \sqrt{\cos x + \dots \infty}}}$ ....

prove that

Q⑧ find $\frac{dy}{dx}$:

⑨ $y = \frac{\sin x + x^2}{\cot 2x}$ ⓑ $y = \tan^{-1}\left(\sqrt{1+x^2} + x\right)$

Q9 If $y = (\sin^{-1} x)^2$ prove that $(1-x^2)$

$$(1-x^2)\frac{d^2y}{dx^2} = x\frac{dy}{dx} + 2$$

Q10 find $\frac{dy}{dx}$ of any two of the following if

(a) $y = \frac{x \tan x}{\sec x + \tan x}$ (b) $y = \sqrt{\sin\sqrt{\sin\sqrt{x}}}$

Q11 If $y = e^x(\sin x + \cos x)$ prove that

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2$$

Q12 prove that $\sqrt{3}\sin x + 3\cos x$ has a maximum value at $x = \pi/6$

Q13 find $\frac{dy}{dx}$ of any two of the following.

(a) $y = \tan^{-1}\left(\sqrt{\frac{1+\cos x}{1-\cos x}}\right)$

Q14 find the maximum and minmum value

$$x^3 + 6x^2 + 9x + 15.$$

Q15 find $\frac{dy}{dx}$ where $y = (\sin x)^x + (\cos x)^{\tan x}$

Q(16) find the maxminum and minima of the function.

$2x^3 - 15x^2 + 36x + 11$

Q(17) If $x = 2\cos\theta - \cos 2\theta$ and $y = 2\sin\theta - \sin 2\theta$ Show that $\frac{dy}{dx} = -1$ at $\theta = \pi/2$

Q(18) If $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ find $\frac{dy}{dx}$

Q(19) If $\sin y = x \sin(a+y)$ then prove that

$$\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$$

Q(20) If $y = (\tan x)^{\cot x}$, find $\frac{dy}{dx}$

Q(21) find the differential coefficient of the function $\sec^{-1} x$ from the first principle.

Q(22) ~~If $\sin y =$~~ find $\frac{dy}{dx}$ of $y = (x)^{\cos x} + (\sin x)^{\tan x}$

Q(23) If $y = (\sin x)^{(\sin x)^{(\sin x)\ldots\infty}}$ Show that

$$\frac{dy}{dx} = \frac{y^2 \cot x}{(1 - y\log_e \sin x)}$$

Q 24. find differential co-efficient of $\operatorname{cosec}^{-1}x$

Q 25. find the maxima and minima of the function $2x^2-4x-8$

Q 26. If $y=\sqrt{\tan x+\sqrt{\tan x+\sqrt{\tan x+\ldots\infty}}}$ prove that $\frac{dy}{dx}=\frac{\sec^2 x}{2y-1}$

Q 27. If $y=e^{x}\log_e(\sin 2x)$ find $\frac{dy}{dx}$

Q 28. If $y=(\tan^{-1}x)^2$ prove that $(1+x^2)^2 y_2+2x(1+x^2)y_1=2$

Q 29. ~~find~~ If $u=e^{xyz}$ show that $\frac{d^3u}{dx\cdot dy\cdot dz}=e^{xyz}\left[1+3xyz+x^2y^2z^2\right]$

Q 30. find the maximum and minimum values of x^3-2x^2+x+6

* find $\frac{dy}{dx} = ?$

(i) $xy = x^3 + y^3$ Ans $3x^2 - y / x - 3y^2$

(ii) $x^m y^n = (x+y)^{m+n}$ Ans y/x

(iii) $x = y \log(xy)$ Ans $y(x-y)/x(x+y)$

(iv) $x^n + y^n = a^n$ Ans $-x^{n-1}/y^{n-1}$

(v) $x^{2/3} + y^{2/3} = a^{2/3}$ Ans $-y^{1/3}/x^{1/3}$

(vi) $y = (x+y)^2$ Ans $2(x+y) / 1-2x-2y$

(vii) $y = \sin(x+y)$ Ans $\cos(x+y) / 1-\cos(x+y)$

(viii) $xy = \sin(x+y)$ Ans $\cos(x+y) - y / x - \cos(x+y)$

(ix) $\log y = \sin(x+y)$ Ans $y\cos(x+y) / 1 - y\cos(x+y)$

(x) $y = \sqrt{x + \sqrt{x + \sqrt{x \ldots \infty}}}$ Ans $1/2y-1$

(xi) $y = \sqrt{\cos x + \sqrt{\cos x + \sqrt{\cos x \cdots \infty}}}$ Ans

(xii) $x^m y^n = (x-y)^{m+n}$ Ans y/x

(xiii) $x/y = \operatorname{cosec} xy$ Ans $\dfrac{\sin(xy) + xy\cos(xy)}{1 - x^2\cos(xy)}$

(xiv) $x^2 y^2 = \sin(xy)$ Ans $-y/x$

(xv) $e^{xy} + xy = 0$ Ans $-y/x$

Q 31 find the differential coefficient of $\sec^{-1}\theta$ from the first principle